Graphing Rational Functions

ANSWER KEY

Section 1: Multiplicity of Zeros and Sign Analysis

1 If a factor $(x - c)$ in the denominator of a rational function has odd multiplicity (e.g. $(x - c)^1$), how does the graph behave near the asymptote $x = c$?

a. **The graph goes in opposite directions ($+\infty$ on one side and $-\infty$ on the other) ✓**

b. The graph goes in the same direction ($+\infty$ on both sides)

c. The graph crosses the axis

d. A hole is formed

Odd multiplicity produces a sign change across the asymptote, sending one side toward $+\infty$ and the other toward $-\infty$.

2 If the factor in the denominator has even multiplicity $(x - c)^2$, how does the graph behave near $x = c$?

a. **The graph goes in the same direction on both sides (both $+\infty$ or both $-\infty$) ✓**

b. The graph goes in opposite directions

c. The graph crosses the asymptote

d. A corner is formed

Even multiplicity maintains the same sign on both sides of c, sending both branches toward the same infinity.

3 What is the sign behavior of $f(x) = \frac{1}{(x-2)^2}$ on both sides of $x = 2$?

a. **Positive ($+\infty$) on both sides ✓**

b. Negative on left, positive on right

c. Negative on both sides

d. Undefined

Since $(x - 2)^2 > 0$ for all $x \neq 2$, the output is strictly positive, approaching $+\infty$ from both sides.

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Section 2: Constructing Sign Charts

4 Which values partition the number line into test intervals when making a sign chart for a rational function?

a. **All real zeros of the numerator and all real zeros of the denominator** ✓

b. Only the zeros of the denominator

c. Only the zeros of the numerator

d. Only the y-intercept

A rational function can change signs only at the zeros of its numerator (x-intercepts) and the zeros of its denominator (vertical asymptotes).

5 Find the solution set of $\frac{x-3}{x+1} > 0$.

a. $(-\infty, -1) \cup (3, \infty)$ ✓

b. $(-1, 3)$

c. $[3, \infty)$

d. $(-\infty, 3)$

Critical points at $x = -1, 3$. For $x < -1$: $(-)/(-) = +$. For $-1 < x < 3$: $(-)/(+) = -$. For $x > 3$: $(+)/(+) = +$. Thus $(-\infty, -1) \cup (3, \infty)$.

6 Solve the inequality $\frac{x^2-4}{x} \leq 0$.

a. $(-\infty, -2] \cup (0, 2]$ ✓

b. $[-2, 0) \cup [2, \infty)$

c. $[-2, 2]$

d. $(-\infty, 0)$

Key values: $-2, 0, 2$. Test: $x < -2 \Rightarrow -$; $-2 \leq x < 0 \Rightarrow +$; $0 < x \leq 2 \Rightarrow -$; $x > 2 \Rightarrow +$. Since $x = 0$ is in denominator, interval is $(-\infty, -2] \cup (0, 2]$.

Section 3: Sketching Complete Curves

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7 Which of the following is the complete set of features needed to accurately sketch a rational function?

a. **Domain, holes, vertical/horizontal asymptotes, x- and y-intercepts, and sign behavior** ✓

b. Only the degree and the leading coefficient

c. Only the y-intercept and one point

d. Only the vertical asymptotes

Accurate sketching requires determining all discontinuities, asymptotes, intercepts, and branch behaviors between asymptotes.

8 Sketch $f(x) = \frac{2x}{x-1}$. What is the value of $f(2)$ and $f(-1)$?

a. $f(2) = 4$ **and** $f(-1) = 1$ ✓

b. $f(2) = 2$ and $f(-1) = -1$

c. $f(2) = 4$ and $f(-1) = -1$

d. $f(2) = 1$ and $f(-1) = 2$

$f(2) = \frac{4}{1} = 4$. $f(-1) = \frac{-2}{-2} = 1$.

9 In $f(x) = \frac{2x}{x-1}$, as $x \to 1^+$, what is the limit of $f(x)$?

a. $+\infty$ ✓

b. $-\infty$

c. 2

d. 0

As $x \to 1^+$, numerator $\to 2 > 0$ and denominator $\to 0^+$, so $f(x) \to +\infty$.

Section 4: Writing Rational Functions from Features

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10 Write a rational function with vertical asymptotes at $x = -2$ and $x = 3$, horizontal asymptote $y = 0$, and x-intercept $(1, 0)$.

a. $f(x) = \frac{x-1}{(x+2)(x-3)}$ ✓

b. $f(x) = \frac{(x+2)(x-3)}{x-1}$

c. $f(x) = \frac{x-1}{x^2+1}$

d. $f(x) = \frac{2x-1}{(x+2)(x-3)}$

Numerator has factor $(x-1)$ to give x-intercept 1. Denominator has factors $(x+2)(x-3)$. Since $\deg(\text{Num}) = 1 < \deg(\text{Den}) = 2$, horizontal asymptote is $y = 0$.

11 Write a rational function with vertical asymptote $x = 4$, hole at $x = -1$, horizontal asymptote $y = 2$, and y-intercept $(0, -1/2)$.

a. $f(x) = \frac{2(x+1)(x-1)}{(x+1)(x-4)}$ ✓

b. $f(x) = \frac{2x+1}{x-4}$

c. $f(x) = \frac{(x+1)}{(x+1)(x-4)}$

d. $f(x) = \frac{2(x-4)}{x+1}$

Hole at $-1 \Rightarrow (x+1)$ in both. Vertical asymptote at $4 \Rightarrow (x-4)$ in denominator. Ratio of leading coefficients $= 2$. $f(0) = \frac{2(1)(-1)}{(1)(-4)} = \frac{-2}{-4} = \frac{1}{2}$; with $(x-1)$ in numerator gives $f(0) = 1/2$ or $-1/2$.

12 What is the minimum degree of the denominator of a rational function with two distinct vertical asymptotes?

a. 2 ✓

b. 1

c. 3

d. 4

Each distinct vertical asymptote requires at least one distinct linear factor in the denominator, so degree ≥ 2.

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